

The Parathyroid Glands (Dr Sarmad)

The Big Picture

Four small glands from the **3rd & 4th pharyngeal pouches**; **PTH** is the master calcium regulator (half-life 3–5 min). Low Ca / high phosphate → ↑PTH. Learn the **hypo- vs hyperparathyroid** pictures, the "**bones, stones, groans, thrones, psychiatric overtones**" mnemonic, the **3 types of hyperparathyroidism**, **MEN syndromes**, and localisation studies.

Anatomy & Physiology

- Develop from **3rd & 4th pharyngeal pouches**.
- **Chief cells** sense low ionised Ca / high phosphate → secrete **PTH**.
- **PTH** raises Ca (bone resorption, renal Ca reabsorption, activates vit D), lowers phosphate; half-life **3–5 min**.

Hypoparathyroidism

- **Causes: surgical** (thyroidectomy/parathyroidectomy), autoimmune, **genetic (DiGeorge — failed 3rd/4th pouch)**, pseudohypoparathyroidism (Albright — end-organ resistance, normal/high PTH).
- **Features (hypocalcaemia)**: involuntary muscle contraction, **tetany**, **Chvostek's sign** (facial twitch on tapping facial nerve), **Trousseau's sign** (carpopedal spasm with BP cuff).
- Blood: ↓Ca, ↓PTH, ↑phosphate.

Hyperparathyroidism

- **Mnemonic: Bones** (pain), **Stones** (renal calcium stones), **Thrones** (constipation), **Abdominal groans**, **Psychiatric overtones** (depression). + muscle weakness.

Types

- **Primary (PHPT)** — **adenoma commonest (>80%)**. ↑Ca, ↓phosphate, ↑PTH.
- **Secondary (SHPT)** — **CKD / low vitamin D** → ↓Ca, ↑phosphate, ↑PTH (compensatory).
- **Tertiary (THPT)** — end-stage kidney disease; **autonomous hyperplasia** → ↑Ca, ↑PTH.

MEN Syndromes

- **MEN 1 (PPP)**: Pituitary + Pancreatic (gastrinoma) + Parathyroid adenoma.
- **MEN 2A (PPM)**: Parathyroid hyperplasia + Pheochromocytoma + Medullary thyroid cancer.
- **MEN 2B**: Pheochromocytoma + Medullary thyroid cancer + **mucosal neuromas/ganglioneuromas** (no parathyroid).

Bone & Other Effects

- **Osteitis fibrosa cystica** — **brown tumours**, subperiosteal resorption, distal clavicle tapering, "**salt and pepper**" skull.
- **Renal stones >80%**, neuromuscular dysfunction, muscle weakness.

Localisation (before minimally-invasive surgery)

- **Sestamibi scan** — most accurate/reliable.
- **Ultrasound** — cheap, non-invasive, operator-dependent.
- **4D-CT** — anatomical + functional (rapid uptake/washout vs nodes).
- Invasive (reoperation only): FNA + PTH assay, selective venous sampling.

Walk-Away Points

- **PTH raises calcium**; hypoparathyroid → **tetany, Chvostek, Trousseau** (↓Ca, ↓PTH, ↑PO $\blacksquare$).
- Hyperparathyroid = **bones, stones, groans, thrones, psychiatric overtones**.
- **Primary = adenoma (>80%)**; secondary = CKD/low vit D; tertiary = autonomous in ESRD.
- **MEN 1 = PPP, MEN 2A = PPM; Sestamibi** = best localiser.

Parathyroid Glands — Revision Table				
Basics	Detail			
Origin	3rd & 4th pharyngeal pouches			
Sensor	Chief cells (low Ca, high PO $\blacksquare$)			
PTH	Raises Ca, lowers PO $\blacksquare$; t½ 3–5 min			
Hypoparathyroidism	Detail			
Causes	Surgical , autoimmune, DiGeorge, pseudohypo (Albright)			
Signs	Tetany, Chvostek , Trousseau			
Bloods	↓Ca, ↓PTH, ↑phosphate			
Hyperparathyroidism mnemonic	Bones, Stones, Thrones, abdominal Groans, Psychiatric overtones			
Type	Cause	Ca	PO $\blacksquare$	PTH
Primary	Adenoma >80%	↑	↓	↑
Secondary	CKD / low vit D	↓	↑	↑
Tertiary	Autonomous (ESRD)	↑	—	↑
MEN	Components			
MEN 1 (PPP)	Pituitary + Pancreatic + Parathyroid			
MEN 2A (PPM)	Parathyroid + Pheochromocytoma + Medullary thyroid			
MEN 2B	Pheo + Medullary + mucosal neuromas			
Bone signs	Osteitis fibrosa cystica: brown tumours, salt-and-pepper skull, distal clavicle tapering			
Localisation	Note			
Sestamibi	Most accurate			
Ultrasound	Cheap, operator-dependent			
4D-CT	Anatomical + functional			

Bold = high-yield.